Finite-Size Scaling of the Two-Dimensional Ising Model with Metropolis, Wolff and CUDA Monte Carlo

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Abstract

Monte Carlo simulations of the ferromagnetic Ising model on $L \times L$ periodic square lattices are used to measure the critical temperature and critical exponents by finite-size scaling. Three simulators are written in C++ and CUDA: single-spin Metropolis on the CPU, the Wolff cluster algorithm, and checkerboard Metropolis on the GPU. All three reproduce exact enumeration of the 4×4 lattice. Every error bar comes from a block bootstrap of the raw Markov chains, so correlated samples are not counted as independent. Wolff chains at the exact critical temperature, for $L = 16$ to 256, give $T_c = 2.26896 \pm 0.00041$ from the Binder-cumulant crossing of the two largest lattices, $\beta/\nu = 0.1266 \pm 0.0016$, $\gamma/\nu = 1.7477 \pm 0.0025$ and $\eta = 0.2523 \pm 0.0025$. All agree with Onsager's solution within one standard deviation. A GPU temperature scan gives a consistent but less precise picture. Earlier results from this project ($T_c = 2.260$, $\beta = 0.096$) were caused by a bug in the Wolff update and by an analysis tuned to the known answer; both are corrected here.

1 Introduction

The two-dimensional Ising model is the standard test case for numerical studies of continuous phase transitions. It has a non-trivial critical point, and Onsager's exact solution [1] gives the critical temperature and exponents to compare against.

A simulation, however, runs on a finite lattice. There the divergences of the infinite system are rounded and shifted, and the critical behaviour appears only through how observables change with L. Finite-size scaling [2, 3] turns that dependence into estimates of T_c and the exponents.

Near T_c a second difficulty appears. Successive Monte Carlo configurations are strongly correlated, so a long run can contain few independent samples, and naive error bars are far too small. This project measures the correlation time of every chain and takes all errors from a block bootstrap.

2 Model and Exact Results

The Hamiltonian is

$$H = -J \sum_{\langle ij \rangle} s_i s_j, \qquad s_i = \pm 1, \tag{1}$$

with the sum over nearest-neighbour pairs and $J = k_B = 1$. Below T_c the spins order and the magnetisation is non-zero; above T_c thermal fluctuations destroy the order.

Onsager's solution gives

$$T_c = \frac{2}{\ln\bigl(1 + \sqrt{2}\bigr)} = 2.269185\ldots, \tag{2}$$

and the exponents $\nu = 1$, $\beta = 1/8$, $\gamma = 7/4$, $\eta = 1/4$ and $\alpha = 0$. The last one means that the specific heat diverges logarithmically rather than as a power law.

3 Observables

All observables are per spin, with $N = L^2$, $m = \frac{1}{N}\sum_i s_i$ and $e = H/N$:

Observable	Definition
Susceptibility (scan)	$\chi' = N\left(\langle m^2\rangle - \langle\lvert m\rvert\rangle^2\right)/T$
Susceptibility at T_c	$\chi = N\langle m^2\rangle / T$
Specific heat	$C = N\left(\langle e^2\rangle - \langle e\rangle^2\right)/T^2$
Binder cumulant	$U = 1 - \langle m^4\rangle / \left(3\langle m^2\rangle^2\right)$

Table 1: Measured observables.

On a finite lattice the sign of m flips during a long run, so $\langle m \rangle = 0$ at every temperature. The scan therefore uses $\langle |m| \rangle$ and the connected susceptibility χ'. At T_c the fixed-temperature analysis uses $\chi = N\langle m^2 \rangle / T$, whose finite-size scaling has no subtracted term.

4 Finite-Size Scaling

At T_c the correlation length of a finite lattice is limited by L, and observables scale as powers of L:

$$\langle |m| \rangle \sim L^{-\beta/\nu}, \qquad \chi \sim L^{\gamma/\nu}, \qquad \chi(T_c) \sim L^{2-\eta}. \tag{3}$$

The exponent ratios are therefore slopes on a log-log plot against L. They need no fitting window in $T_c - T$, which was the weak point of the earlier analysis (Section 9).

Near T_c the Binder cumulants $U_L(T)$ of different sizes cross at a common point. Its temperature approaches T_c as L grows, and its height approaches the universal value $U^* = 0.61069$ [4]. Away from T_c, the susceptibility peak shifts as

$$T_c(L) = T_c + aL^{-1/\nu}. \tag{4}$$

5 Monte Carlo Algorithms

5.1 Three simulators

Binary	Update	One "sweep"	Notes
`ising2d_cpu`	Metropolis, checkerboard order	N attempted flips	reference implementation
`ising2d_wolff`	Wolff single cluster	one cluster flip	bonds added with $p = 1 - e^{-2/T}$
`ising2d_cuda`	Metropolis, checkerboard, one thread per site	N attempted flips	`curand` XORWOW per thread; needs even L

Table 2: Simulators.

Metropolis accepts a single-spin flip with probability $\min(1, e^{-\Delta E/T})$. The checkerboard order updates all sites of one sublattice at once, which is what makes the GPU version possible. With odd L, periodic neighbours would share a colour, so the CUDA code rejects odd lattices.

The Wolff algorithm [5] grows a cluster from a random seed, adding each aligned neighbour with probability $p = 1 - e^{-2/T}$, and flips the whole cluster. Near T_c clusters

are large and the chain decorrelates in a few flips. Far below T_c clusters cover most of the lattice, and the algorithm is slow.

All drivers start from the ordered state. A hot start below T_c can freeze Metropolis into a striped two-domain state on large lattices.

5.2 Validation against exact enumeration

The 4×4 lattice has 2^{16} states, few enough to sum exactly. `make check` runs each built simulator at $T = 1.5$, T_c and 3.0 and compares $\langle e \rangle$ and $\langle |m| \rangle$ with the exact values. A check fails above five standard errors. All 18 checks pass. The CUDA build is also clean under `compute-sanitizer` memcheck and racecheck.

6 Statistical Analysis

6.1 Autocorrelation and block bootstrap

Successive samples of a Markov chain are correlated over an integrated autocorrelation time τ_{int}, estimated with Sokal's windowing ($W \geq 6\tau$). A chain of n samples holds only about $n/2\tau_{\mathrm{int}}$ independent ones. For Metropolis at T_c, $\tau_{\mathrm{int}} \approx 1000$ sweeps at $L = 128$ and about 6000 at $L = 256$, so 200 000 sweeps contain only about 100 and 16 independent samples. Wolff reduces this to a few cluster flips.

Errors therefore come from a block bootstrap. Each chain is cut into 50 contiguous blocks, each much longer than τ_{int}, and the blocks are resampled with replacement 200 times. Every quantity, including fitted slopes and crossings, is recomputed on each replicate.

6.2 Single-histogram reweighting

A chain at T_0 predicts averages at nearby T [6]:

$$\langle A \rangle_T = \frac{\sum_t A_t \, e^{-(1/T - 1/T_0)E_t}}{\sum_t e^{-(1/T - 1/T_0)E_t}}, \tag{5}$$

where E_t is the total energy of sample t. The usable range is about $|T - T_0| \lesssim T_0^2/\sigma_E$, which is ± 0.005 at $L = 256$. Each Binder crossing is searched for only inside that window.

7 Fixed-Temperature Finite-Size Scaling

7.1 Runs

One Wolff chain per size runs at the exact T_c for $L = 16, 32, 64, 128, 256$, with 10 000 thermalisation flips and 60 000 measured flips (`make fss`, 72 s on one core). The raw time series of m and e are saved. The bootstrap blocks are 99 to 467 integrated autocorrelation times long.

7.2 Results

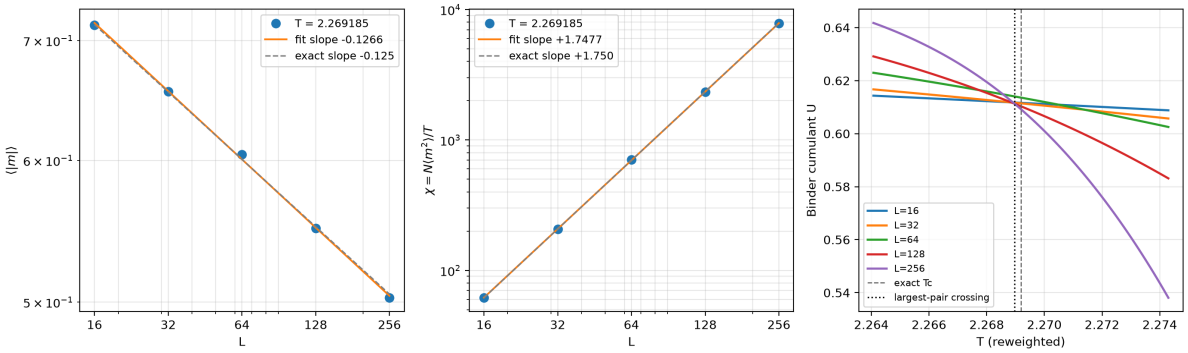


Figure 1: Fixed-temperature finite-size scaling. Left and centre: $\langle|m|\rangle$ and $\chi = N\langle m^2\rangle/T$ at the exact T_c against L, with the fitted (solid) and exact (dashed) slopes. Right: Binder cumulants reweighted to nearby temperatures; the dotted line is the crossing of $L = 128$ and 256, the dashed line the exact T_c.

Quantity	Result	Exact	Deviation
T_c, Binder crossing $L = 128/256$	2.26896 ± 0.00041	2.269185	0.5σ
β/ν at exact T_c	0.1266 ± 0.0016	1/8	1.0σ
γ/ν at exact T_c	1.7477 ± 0.0025	7/4	0.9σ
$\eta = 2 - \gamma/\nu$ at exact T_c	0.2523 ± 0.0025	1/4	0.9σ
β/ν at measured T_c	0.1242 ± 0.0042	1/8	0.2σ
γ/ν at measured T_c	1.7517 ± 0.0070	7/4	0.3σ

Table 3: Fixed-temperature results. Errors are one standard deviation from the block bootstrap.

The rows "at measured T_c" reweight each chain to this run's own Binder crossing, and every bootstrap replicate recomputes that crossing. Those exponents do not use the exact T_c at all, and their errors include the uncertainty of T_c. With $\nu = 1$, exact in two dimensions, the measured-T_c value gives $\beta = 0.1242 \pm 0.0042$.

L	16	32	64	128	256
$U(T_c)$	0.6116	0.6114	0.6136	0.6103	0.6092

Table 4: Binder cumulant at the exact T_c, each ± 0.001, against $U^* = 0.61069$.

The Binder crossings of adjacent sizes are 2.2688 ± 0.0022 (16/32), 2.2714 ± 0.0016 (32/64), 2.2677 ± 0.0007 (64/128) and 2.26896 ± 0.00041 (128/256). Only the largest pair is quoted as the result, because corrections to scaling are smallest there.

All 18 quantities agree with the exact values within about 2.5σ. The largest deviation is $U(T_c)$ at $L = 64$ (2.6σ). The same chain explains why the 32/64 and 64/128 crossings fall on opposite sides of T_c. With 18 quantities, one value at that level is not unusual.

8 GPU Temperature Scan

8.1 Runs

The CUDA Metropolis simulator scans $T = 2.0$ to 2.6 at 76 temperatures, with steps of 0.005 inside $[2.1, 2.4)$, for $L = 16$ to 256. Each point uses 5000 thermalisation sweeps and 200 000 measured sweeps, sampled every 50 sweeps. The scan stores one chain per (T, L) and no time series, so its errors are regression errors only. It gives an overview of the transition, not the quoted numbers.

8.2 Thermodynamic curves

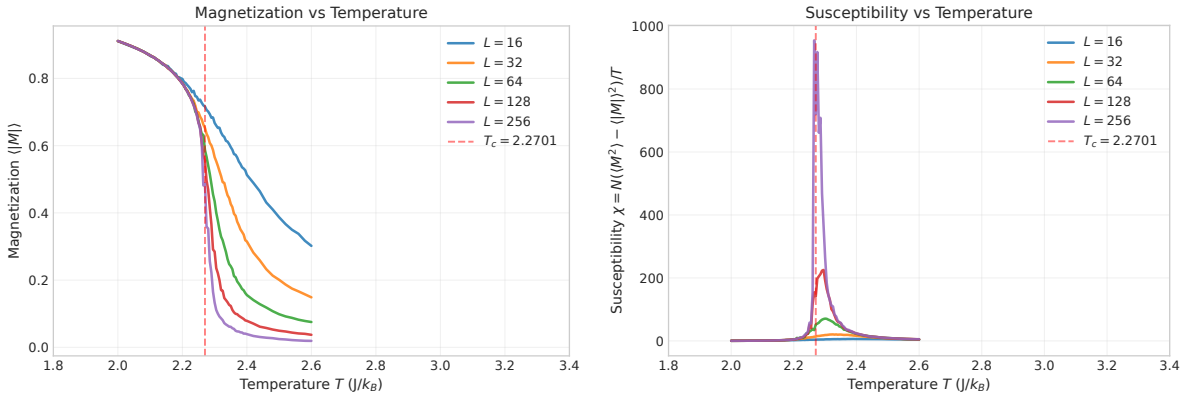


Figure 2: Magnetisation $\langle |m| \rangle$ (left) and susceptibility χ' (right) against temperature. The dashed line is the scan's extrapolated $T_c = 2.2701$.

The magnetisation falls more steeply with increasing L, as the rounded finite-size transition approaches a true singularity. The susceptibility peak grows from 6.3 at $L = 16$ to 954 at $L = 256$ and moves towards T_c. At $L = 256$ the curve is noisy near the peak, because each point holds only about 16 independent samples.

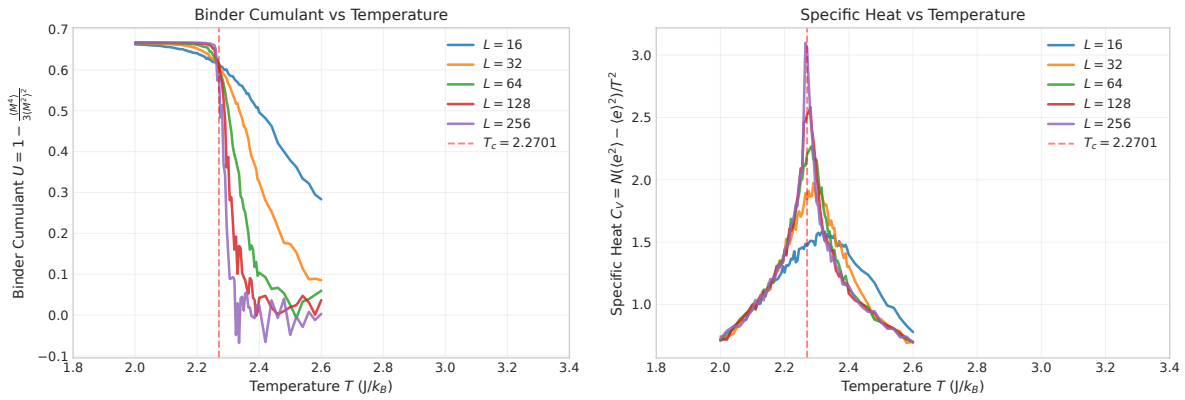


Figure 3: Binder cumulant (left) and specific heat (right) against temperature.

The Binder cumulant drops from 2/3 in the ordered phase towards 0 in the disordered phase, more sharply for larger L. The specific heat peak grows slowly with L. A logarithmic divergence ($\alpha = 0$) predicts $C_{\max} = a + b \ln L$ with $b = 8K_c^2/\pi = 0.4945$ for the exact $K_c = 1/T_c$. The scan gives $b = 0.487 \pm 0.011$.

8.3 Critical temperature and exponents

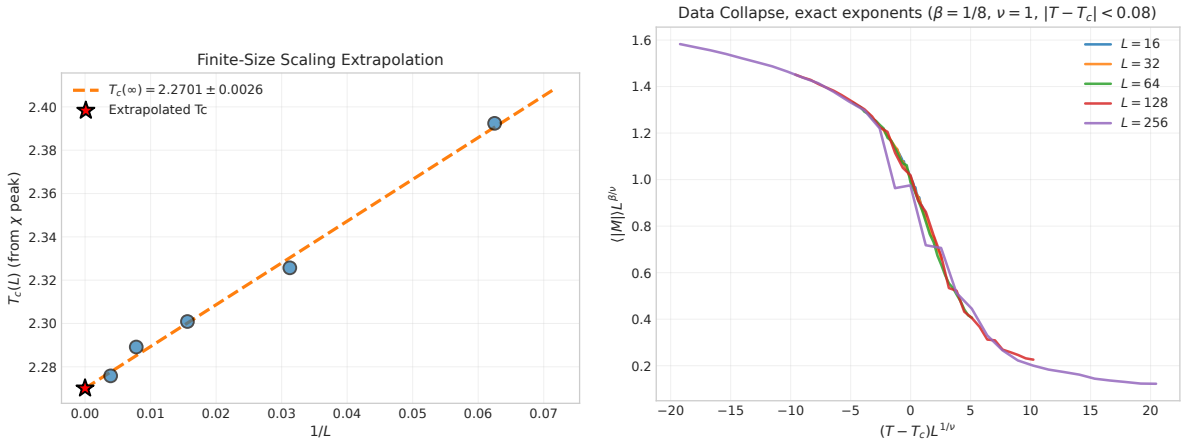


Figure 4: Left: $T_c(L)$ from the susceptibility peaks against $1/L$, with the extrapolation to $L \to \infty$. Right: data collapse of $\langle |m| \rangle L^{\beta/\nu}$ against $(T - T_c)L^{1/\nu}$ with the exact exponents.

Quantity	Scan result	Exact
T_c, χ'-peak extrapolation	2.2701 ± 0.0026	2.269185
T_c, Binder crossing $L = 32/64$	2.2681 (no error estimate)	2.269185
γ/ν from $\chi'_{\max} \sim L^{\gamma/\nu}$	1.746 ± 0.016	7/4
η from $\chi(T_c) \sim L^{2-\eta}$	0.262 ± 0.010	1/4
$C_{\max}$ against $\ln L$, slope	0.487 ± 0.011	$8K_c^2/\pi = 0.4945$

Table 5: Temperature-scan results. Errors are regression errors only.

The peak positions $T_c(L) = 2.3924$, 2.3257, 2.3009, 2.2891 and 2.2758 for $L = 16$ to 256 lie close to a straight line in $1/L$, as $\nu = 1$ predicts, and extrapolate to 2.2701 ± 0.0026. With the exact exponents the magnetisation curves of all sizes collapse onto one function. The scatter of the $L = 256$ curve near T_c is the same sampling noise as in Fig. 2.

The scan's η depends strongly on T_c: moving T_c by its standard error shifts η by 0.039. The fixed-temperature value (0.2523 ± 0.0025) should be quoted instead. The scan also rejects the 64/128 and 128/256 Binder crossings as ambiguous: with about 16 independent samples per point at $L = 256$, the cumulant curves cross several times by noise.

8.4 Parameters

Parameter	Fixed-T scaling	Temperature scan
Algorithm	Wolff (CPU)	Metropolis (CUDA)
Lattice sizes	$16, 32, 64, 128, 256$	$16, 32, 64, 128, 256$
Temperatures	exact T_c	76 points in $[2.0, 2.6]$
Thermalisation	10 000 cluster flips	5000 sweeps
Measurement	60 000 cluster flips, every flip	200 000 sweeps, every 50
Errors	block bootstrap, 200×50 blocks	regression only

Table 6: Simulation parameters.

9 Corrections to the Earlier Analysis

An earlier version of this project reported $T_c = 2.260 \pm 0.003$ and $\beta = 0.096 \pm 0.002$. Both are superseded, for the following reasons.

- **Wolff bug.** A neighbour that failed one bond test was marked as visited and could never join the cluster through another bond. On the 4×4 lattice at $T = 1.5$ this gave $e = -0.83$ instead of -1.95, so all earlier Wolff data was invalid. The Metropolis paths were not affected. The exact-enumeration check now catches this kind of error.

- $\beta = 0.096$ **came from the** T_c **estimate.** A log-log fit of $\langle|m|\rangle$ against $T_c - T$ turns a 0.4% error in T_c into a 24% error in β. It is replaced by L-scaling at fixed T, which needs no fitting window.

- **The** T_c **pipeline was tuned to the known answer.** It dropped sizes below 64, forced $T_c(L)$ to decrease with L, fell back whenever the result left the range 2.1 to 2.4, weighted the fit by L^2 and smoothed U with a Savitzky–Golay filter. All of this is removed.

- **The specific heat was missing a factor of** N, so C shrank with L.

- **Errors ignored autocorrelation.** Naive errors were about six times too small at $L = 128$.
- **The CUDA kernel read past the random-number array for odd L.** Odd L is now rejected.

10 Limitations

- The scan errors are regression errors only, because the scan stores no time series. Rerunning it with `--series-dir` gives block-bootstrap errors for the Binder cumulant.
- Metropolis is too slow at large L near T_c: $\tau_{\text{int}} \propto L^z$ with $z \approx 2.17$, so the $L = 256$ scan has about 16 independent samples per point. Wolff near T_c, or more sweeps over a coarser critical band, are the practical fixes.
- $\nu = 1$ is assumed, not fitted, in the $T_c(L)$ extrapolation and the data collapse. The fixed-temperature analysis measures only the ratios β/ν and γ/ν.
- Corrections to scaling are not fitted. Dropping $L = 16$ moves β/ν from 0.1266 to 0.1286 ± 0.0024, within errors.

11 Conclusion

Three Monte Carlo simulators of the 2D Ising model pass exact enumeration on the 4×4 lattice. Wolff chains at the exact critical temperature give $T_c = 2.26896 \pm 0.00041$, $\beta/\nu = 0.1266 \pm 0.0016$, $\gamma/\nu = 1.7477 \pm 0.0025$ and $\eta = 0.2523 \pm 0.0025$, all within one standard deviation of Onsager's values. The errors are realistic because they come from a block bootstrap of the raw chains. The earlier 23% discrepancy in β was not a finite-size effect: it came from a Wolff bug and from an exponent fit that amplified a small error in T_c.

References

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